Many Daughters
Pre-Analysis Plan

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1 Analysis Plan

In the following, we describe the specifications with which we test each of the four hypotheses. For all tests, we rely on linear OLS regressions with robust standard errors, and we use two-sided hypothesis tests with $p < .05$ as our significance threshold.

1.1 Identification Strategy

Our primary independent variable is whether the respondent's first child was a boy or a girl. We consider only the first birth, as the sex of the first child is random, while any subsequent fertility decisions might be influenced by the sex of the first child. Hence, comparing respondents whose first child was a boy to respondents whose first child was a girl gives us the causal impact of the sex of the first child.

In testing all four hypotheses, we always use the same two control variables: the respondent's gender and birth year (for H3, which is defined at the household level, we do not control for gender but control for the birth year of the male and the female household member). We expect these controls to explain some of the residual variation in our outcome measures, thereby increasing the power of the tests. At the same time, we are confident that these control variables are not endogenous to the gender of the first child. That is, we refrain from including variables such as the respondent's occupation, income, education, or the number children, as these may be influenced by the gender of the first child.

The analysis plan is structured as follows: we detail the variables and specifications we use for each of the hypotheses in Sections 1.2-1.5. In Section 1.6, we report additional exclusion criteria. A list of the variables used and their descriptions are included in Section 1.7.

1.2 Hypothesis 1

The first hypothesis is as follows:

Hypothesis 1 *Having daughters leads to stronger support for gender equality (as measured by attitudes). That is, denoting X as a measure of support for gender equality,*
(X for respondents with daughters) > (X for respondents without daughters).

To test this hypothesis, we first construct a measure of 'support for gender equality'. To do this, we calculate an index of responses to six questions that measure attitudes towards different aspects of gender quality:

- Men and women should have equal democratic rights.
- Working is the best way for women to be independent.
- Even married women should work to be independent.
- It is problematic that a woman earns more than her partner (reverse-coded).
- For parents, the education of sons should be more important than the education for daughters (reverse-coded).
- The husband should have the final say in the home (reverse-coded).

Because these variables are not all measured on the same scale, we first standardise each item individually as a z-score (mean of 0, standard deviation of 1). Then, we sum all variables and standardise the summation to have a final index with a mean of 0 and a standard deviation of 1. In doing so, we set any missing variables to 0, and we exclude observations that answer less than 3 of the 6 questions.

Using our index of gender equality attitudes, we estimate the following OLS regression:

$$\begin{aligned} GenderAttitudes &= \beta_0 + \beta_1 \cdot Male + \beta_2 \cdot BirthYear + \beta_3 \cdot Daughter \\ \text{Test: } &\beta_3 = 0, \end{aligned} \tag{1}$$

where $GenderAttitudes$ is the index created from the six attitudinal variables as explained above, $Daughter$ is a dummy variable equal to 1 if the respondent's first child is a girl, $Male$ is a dummy for whether the respondent is a male, and $BirthYear$ is the respondent's year of birth. Our coefficient of interest is β_3.

1.3 Hypothesis 2

The second hypothesis is as follows:

Hypothesis 2 *Having daughters leads to more left-leaning political preferences. That is, denoting Y as a measure of left-leaning political preferences,*
(Y for respondents with daughters) > (Y for respondents without daughters).

To test this hypothesis, we rely on the respondents' self-reported attitudes on a political left-right scale. To ensure comparability with the specification used to test H1, we first standardise the political attitudes as a z-score (mean of 0, standard deviation of 1). We then estimate the following OLS regression:

$$\begin{aligned} PoliticalAttitudes &= \beta_0 + \beta_1 \cdot Male + \beta_2 \cdot BirthYear + \beta_3 \cdot Daughter \\ \text{Test: } &\beta_3 = 0, \end{aligned} \tag{2}$$

where $PoliticalAttitudes$ is the respondent's self-reported placement on the political left-right scale (reversed such that higher values indicate placement more to the left), $Daughter$ is a dummy variable equal to 1 if the respondent's first child is a girl, $Male$ is a dummy for whether the respondent is a male, and $BirthYear$ is the respondent's year of birth. Our coefficient of interest is β_3.

Because the question about political attitudes was asked in more than one wave of the SOEP Panel, we have more than one outcome for some respondents. We deal with this by including only survey responses recorded after the birth of the first child and by clustering on respondent level.

1.4 Hypothesis 3

The third hypothesis is as follows:

Hypothesis 3 *Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes). That is, denoting Z as a measure of gender equality in the household,*
(Z for respondents with daughters) > (Z for respondents without daughters).

To examine behavioural measures of gender equality in the household, we look at the gross (pretax) income in households with one male and one female adult with at least one child. In particular, we calculate the share of the gross income in the household that the woman earns: denote by $y_{m,h}$ and $y_{f,h}$ the gross income of the male and female member of

household h (matched through the partner's person number). Then, our outcome variable for the household is defined as

$$IncomeShare_h = \frac{y_{f,h}}{y_{m,h} + y_{f,h}} \tag{3}$$

To test the hypothesis, we estimate the following OLS regression using only households for whom the $IncomeShare$ could be calculated:

$$\begin{aligned} IncomeShare_h &= \beta_0 + \beta_1 \cdot BirthYear_{f,h} + \beta_2 \cdot BirthYear_{m,h} + \beta_3 \cdot Daughter_h \\ \text{Test: } &\beta_3 = 0, \end{aligned} \tag{4}$$

where $IncomeShare_h$ is the share of the gross income defined above, $Daughter_h$ is a dummy variable equal to 1 if the household's first child is a girl, and $BirthYear_{f,h}$ ($BirthYear_{m,h}$) is the birth year of the woman (man) in the household.

Because we have data on incomes in more than one wave of the SOEP Panel, we have more than one outcome for some respondents. We deal with this by including only observations recorded after the birth of the first child and by clustering on household level. In practice we ensure that we have only one observation per household per survey year by estimating the regression only for female respondents (the number of observations is the same if we use only male respondents because the outcome variable is only defined for partnerships with one male and one female member).

1.5 Hypothesis 4

The final hypothesis is as follows:

Hypothesis 4 *Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes). That is, denoting X as a measure of support for gender equality,*
(X for male respondents with daughters - X for male respondents without daughters) > (X for female respondents with daughters - X for female respondents without daughters)

We test H4 using an approach analogous to the one we apply for testing H1, the only difference being that we add an interaction with the gender of the respondent. That is, we estimate the following OLS regression:

$$\begin{aligned} GenderAttitudes &= \beta_0 + \beta_1 \cdot Male + \beta_2 \cdot BirthYear + \beta_3 \cdot Daughter + \beta_4 \cdot Daughter \cdot Male \\ \text{Test: } &\beta_4 = 0, \end{aligned} \tag{5}$$

where $GenderAttitudes$ is the index created from the six attitudinal variables as explained above, $Daughter$ is a dummy variable equal to 1 if the respondent's first child is a girl, $Male$

is a dummy for whether the respondent is a male, and $BirthYear$ is the respondent's year of birth. As this hypothesis concerns only differential effects on women and men, our coefficient of interest is β_4.

1.6 Exclusion Criteria

In testing all hypotheses, we rely on the comparison between respondents whose first child was a girl and respondents whose first child was a boy. For this reason, we exclude in our sample (i) all respondents who do not have any children, (ii) respondents who did not specify the gender of their first child, (iii) respondents who did not specify the birth year of their first child, (iv) respondents whose first children were twins, triplets, quadruplets, etc.,[1] and (v) respondents for whom the outcomes were measured before their first child was born.

[1]Because we do not have access to the biotwin data set of the SOEP Panel in which this is listed explicitly, we exclude respondents whose two first children were born in the same year and in the same month, noting that we might lose a few participants who did not report data on the birth month of their child.

1.7 Variable List

Variable	Name	Measurement	Wording
Primary independent variable			
Sex of the first-born child	kidsex01	Female/male	.
Control variables			
Sex of the respondent	sex	Female/male	.
Birth year	gebjahr	Year	.
Outcomes, H1 & H4			
Democratic rights	plj0620	11-point scale	"Women have the same rights as men."
Women working	plj0621	7-point scale	"Having a job is the best way for a woman to be independent."
Married women working	plj0622	7-point scale	"Even a married woman should have a paid job so that she can be financially independent."
Women earning more than partner is problematic	plj0623	7-point scale	"If a woman earns more money than her partner, this inevitably leads to problems."
Education of sons more important than daughters	plj0624	7-point scale	"For parents, vocational training or higher education for their sons should be more important than vocational training or higher education for their daughters."
Husband final say	plj0625	7-point scale	"At home, the husband should have the final say."
Outcomes, H2			
Political attitudes	plh0004	11-point scale	"In politics people often talk about 'left' and 'right' when it comes to characterize different political attitudes. If you think about your own political views: Where would you place yours?"
Outcomes, H3			
Gross income	pglabgro	Euro	"In politics people often talk about 'left' and 'right' when it comes to characterize different political attitudes. If you think about your own political views: Where would you place yours?"

2 Preliminary Results

	Hypothesis 1	Hypothesis 2	Hypothesis 3	Hypothesis 4
Coefficient of interest	-0.100	0.097	-0.023	0.034
Standard error	0.133	0.051	0.025	0.270
p-value	0.452	0.059	0.361	0.899
Significant support for H	No	No	No	No

The preliminary results do not show support for any of the four hypotheses at the chosen significance threshold. The coefficients for hypotheses 2 and 4 are in the expected direction, and the estimate for hypothesis 2 is borderline significant.